Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam A · Form 2: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. A student marks a taut rope on each side of a fixed pulley. In one second the left mass moves up by d. Which observation would contradict the ideal fixed-rope model?

(A) Both travel equal distances

(B) The left mass moves upward at changing speed

(C) The right mass descends by 2d

(D) The right mass descends by d

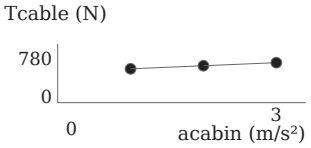
2. Release a 60 kg cabin and adjustable counterweight over an ideal pulley without a motor. Cabin up is positive; g=10 m/s². acabin (m/s²) -> Tcable (N): (1, 660); (2, 720); (3, 780). As acabin (m/s²) increases, what happens to Tcable (N)?

(A) no change

(B) decrease

(C) increase

(D) reversal midway



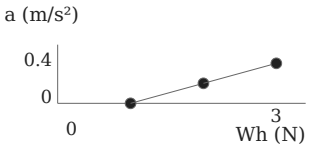
3. A horizontal cart slides toward an ideal pulley. Cart plus hanger mass is fixed at 5 kg; a constant 1 N resistance acts on the cart. Use g=10 m/s². Wh (N) -> a (m/s²): (1, 0); (2, 0.2); (3, 0.4). What is the mean of the first two vertical readings?

(A) 0.1

(B) 0.2

(C) 0.05

(D) 0.15



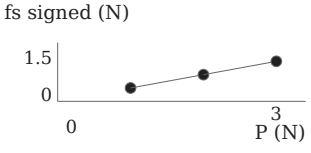
4. An incline block of weight 10 N rests on a 30° ramp with a 5 N hanger on an ideal pulley. Add vertical downward force P to the block. Friction is sufficient; uphill is positive. P (N) -> fs signed (N): (1, 0.5); (2, 1); (3, 1.5). What is the signed change in the vertical reading from first to last trial?

(A) 1.8

(B) 1.4

(C) 0.6

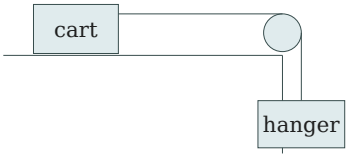
(D) 1



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A frictionless cart and a hanging mass share a taut rope over a fixed pulley. The hanger descends 0.12 m.

a. Mark both displacement directions on the apparatus. **(1 pt)**



b. Use rope length to determine cart travel and compare the average speeds over that interval. **(3 pt)**

6. Transfer mass between a frictionless cart and its hanger on an ideal pulley. Total mass is 5 kg; g=10 m/s². mh (kg) -> a (m/s²): (0.12, 0.24); (0.24, 0.48); (0.36, 0.72).

a. Build the model for track and hanger from physics and predict both slope and intercept. **(2 pt)**

b. Plot the first and last measurements and describe the trend. **(2 pt)**

7. A block on a smooth table is tied to an equal hanging mass m over a fixed ideal pulley. Derive the acceleration in terms of g. **(3 pt)**